

Full length article

Effect of powder feeding rate on heat and mass transfer behaviors during filler powder laser welding

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HIGHLIGHTS

- The effect of powder feeding rate on plasma was investigated.
- The curve of inverse bremsstrahlung absorption and keyhole depth was obtained.
- The reason for the variation in keyhole area was explained by calculating plasma pressure.
- The curve of powder feeding rate and molten pool temperature is similar to a linear line.

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 Heat and mass transfer behavior
 Plasma pressure
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ABSTRACT

The heat and mass transfer behavior is the influencing factor of welding quality, during filler powder laser welding. The characteristics of plasma and keyhole can characterize the heat and mass transfer behavior. Experimental platforms were built to observe the effect of powder feeding rate on the heat and mass transfer behavior. The main results of the experiment are as follows. In the process of filler powder laser welding, the keyhole depth and fusion depth were improved by selecting the critical value of powder feeding rate 5 g/min as the welding process parameter. When the powder feeding rate exceeded the critical value, the electron temperature and electron density decreased and increased respectively. The keyhole depth decreased with the increase of the inverse bremsstrahlung absorption coefficient of the keyhole plasma at the top of the keyhole. The keyhole area increased with the increase of plasma pressure at the keyhole opening. The inverse bremsstrahlung absorption of plasma plume and the number of powder particles sprayed into the molten pool were the factors that caused the decrease of the surface temperature and the length of the molten pool.

1. Introduction

Filler powder laser welding (FPLW) is a new technology of laser welding. The working principle of FPLW is as follows. When the laser beam irradiates the welding base material, the powder particles are transferred to the surface of the welding base material by the powder feeding system. Therefore, FPLW not only inherits the advantages of traditional laser welding, but also can improve the microstructure and mechanical properties of welding seams by changing the chemical composition of powder particles and powder feeding rate. In addition, the gap allowance of FPLW plates is higher than that of traditional laser welding plates. With the increase of powder feeding rate in filler powder laser welding, the gap allowance between plates increases correspondingly. In this paper, the effect of powder feeding rate on heat and mass transfer behavior was studied. Heat and mass transfer

behavior can be characterized by physical quantities in the welding process, such as electron temperature, electron density, keyhole area, keyhole depth, molten pool temperature and length.

The electron temperature and electron density of laser induced plasma are the microscopic heat and mass transfer behaviors during FPLW. The heat and mass transfer behavior of plasma is closely related to welding quality. The welding quality and the formation of welding defects was reflected in real time by the electron temperature of plasma plume [1]. Laser induced plasma can reduce the efficiency of welding. The influence parameters of plasma are laser power density and welding speed. Therefore, the influence of laser welding parameters, such as laser power, welding speed and so on, on the heat and mass transfer behavior of plasma has become a research hotspot. The results can provide appropriate parameters for welding process and improve the quality of welding seam.

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In the past papers, scholars have studied the influence of different laser welding process parameters on the heat and mass transfer behavior of plasma, to further explained the welding quality and the causes of welding defects. Jay et al. [2] measured the electron temperature and electron density of plasma plume and keyhole plasma by adjusting the angle between the optical axis of photodiode and the horizontal plane. The laser energy absorbed by the plasma mainly occurs at the top of the keyhole, which led to a wide upper section of the weld seam. Liu et al. [3] studied the interaction between laser and arc plasma during laser-arc hybrid welding. Experimental results show that the two situations can increase the electron temperature and reduce the electron density. Situation 1 is the process of increasing arc current, and situation 2 is the process of arc plasma passing through the laser beam. Hua et al. [4] studied the plasma behavior in CO₂ laser + GMAW-P hybrid welding process. The electron density and electron temperature distribution showed bimodal behavior in the case of arc leading mode, while this phenomenon does not exist in the case of laser leading mode. Cai et al. [5] mentioned that with the increase of He volume fraction in the Ar-He mixed gas, the shielding effect of plasma is weakened by the decrease of electron temperature and electron density. The above phenomenon eventually led to the reduction of porosity defects. Tao et al. [6] proposed that the reduction of ambient air pressure during laser welding negatively affected the electron temperature and electron density, and positively affected the inverse bremsstrahlung absorption, which positively affected the aspect ratio of the weld section. Luo et al. [7] proposed that helium, as a shielding gas in welding process, can reduce electron temperature and electron density and further inhibited the inverse bremsstrahlung absorption of plasma compared with argon gas. Therefore, helium gas can increase the laser energy absorption efficiency of the material, which can improve the welding efficiency. In addition, the increase of the flow rate of shielding gas can effectively inhibit the plasma eruption by reducing the electron temperature and electron density. Chen et al. [8] studied the effect of the distance (D_{la}) between the optical axis of the laser beam and the welding point of the electrode on the plasma. With the increase of D_{la}, the electron temperature decreased first and then increased. The electron density increased and then decreased. Wang et al. [9] mentioned that laser power directly affected electron temperature and electron density. Different materials corresponded to different turning points of laser power. When the laser power exceeded the turning point, the plasma changed from weak ionization state to strong ionization state. Chen et al. [10] proposed that the plasma distribution in the keyhole was not uniform. Compared with CO₂ laser welding, the electron temperature of fiber laser welding was lower. Jin et al. [11] observed the keyhole plasma characteristics during laser deep fusion welding of 6016 aluminum alloy. With the increase of laser power, the electron temperature of keyhole plasma increased. The electron temperature of keyhole plasma decreased with the increase of welding speed and focal length. Li et al. [12] mentioned that the electron temperature of plasma decreased as the laser pulse duration increased. However, in the case of variations in laser welding parameters, the reasons for variations in electron temperature and electron density have rarely been discussed in the past papers. In addition, the effect of powder feeding rate on the heat and mass transfer behavior of plasma has not been studied.

The keyhole morphology is the macroscopic heat and mass transfer behavior in laser welding. Chen et al. [13] proposed that the prolonged existence time of keyhole outlet could inhibit the porosity, and the larger the keyhole depth/width ratio, the higher the porosity. Zhang et al. [14] studied the keyhole profile morphology and the formation of weld defects. Numerical analysis and experimental results showed that the keyhole width reached quasi-steady state earlier than the keyhole depth. The keyhole depth/width ratio increased, leading to keyhole capillary instability and keyhole collapse. Jiang et al. [15] observed the profile morphology of the keyhole and founded that the quality of the welding seam surface and the depth of fusion increased with the

decrease of ambient air pressure. Under the welding environment of 101 KPa atmospheric pressure, the keyhole oscillated and collapsed periodically. However, in a vacuum welding environment, the keyhole remained stable. Therefore, the welding quality is closely related to the keyhole morphology. Furthermore, the monitoring technology of keyhole morphology has gradually attracted the attention of scholars. Kim et al. [16] designed a coaxial monitoring system, which can capture the keyhole morphology. Through threshold segmentation and edge detection, the keyhole area can be calculated from the acquired image. Zhang et al. [17] proposed the “sandwich” method and built an experimental observation platform to observe the profile morphology of the keyhole.

The temperature and morphology of molten pool is another kind of macroscopic heat and mass transfer behavior in laser welding process. The morphological characteristics of the molten pool can reflect the welding quality in real time [18]. Yin et al. [19] studied the effect of the gap between face plate and core plate on the length of the molten pool. As the gap between plates increased, the length of molten pool decreased. Asghar et al. [20] mentioned that welding speed and laser power have positive effect on molten pool length. The increase of laser power or the decrease of laser beam size led to the increase of the highest temperature of the molten pool. Ashish et al. [21] proposed that as the welding speed increased, the area of molten pool decreased. With the increase of laser power, the molten pool area increased. Yung et al. [22] mentioned that the increase of laser power and the decrease of welding speed led to the increase of molten pool temperature. Nevertheless, the effect of powder feeding rate on macroscopic heat and mass transfer behavior in laser welding has not been discussed. In addition, the relationship between macroscopic and microscopic heat and mass transfer behavior with the change of powder feeding rate has not been studied.

In this paper, based on the “sandwich” method, experimental platforms were built to observe the heat and mass transfer behavior during filler powder laser welding. With the increase of powder feeding rate, the variations of heat and mass transfer behaviors were studied. For the correlation between plasma and keyhole morphology, on the one hand, the reason of the change of keyhole area was explained by the change of plasma pressure at the keyhole opening. On the other hand, the reason for the change of the keyhole depth was explained by the inverse bremsstrahlung absorption of the keyhole plasma at the top of the keyhole. In addition, the mechanism of the effect of powder feeding rate on the keyhole depth was studied, which provided an appropriate powder feeding rate for FPLW to improve the depth of fusion of the welding seam.

2. Experimental

2.1. Experimental platform and apparatus

In this paper, the observation platform of surface heat and mass transfer behavior was built to observe the influence of powder feeding rate on keyhole area, the length and surface temperature of molten pool, and electron temperature and electron density of plasma plume, as shown in Fig. 1. The observation platform of the internal heat and mass transfer behavior was built to observe the effect of powder feeding rate on the keyhole depth, electron temperature and electron density of the keyhole plasma, as shown in Fig. 2. The experimental instruments of the experimental platform were listed as follows. MEMRECAM HX-7 high-speed camera, auxiliary light source, HR2000+ high-resolution spectrometer, IGAR12-LO colorimetric pyrometer, displacement platform. High-speed camera was used to capture the surface and profile morphology of the keyhole. Before the experiment, a neutral density filter and a narrow band-pass filter were installed on the lens at the front of the high-speed camera, which protects the high speed camera and overcomes the intense light interference. The colorimetric pyrometer was used to measure the surface temperature of the molten

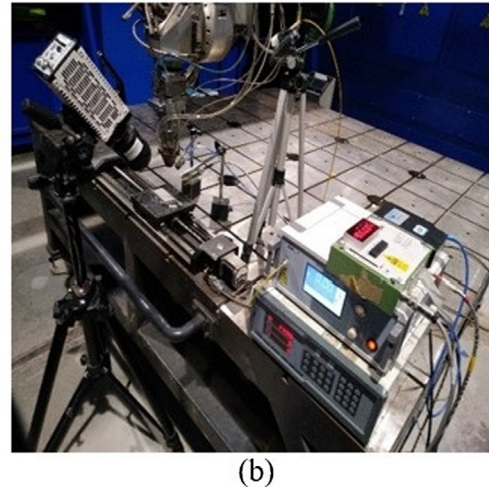
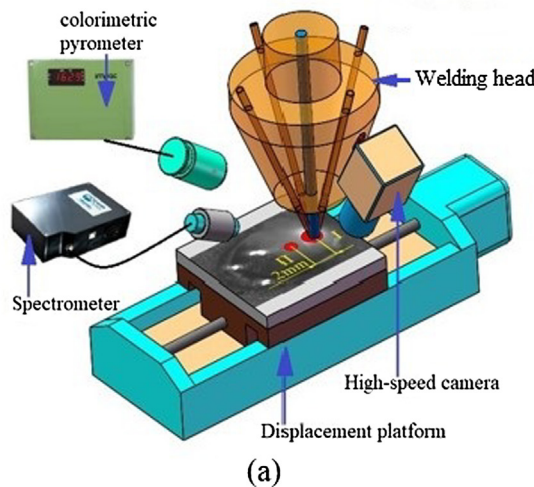


Fig. 1. Surface heat and mass transfer behavior observation platform (a) schematic illustration of experimental set-up (b) real experimental set-up.

pool. Spectrometer was used to measure the spectrum of the plasma plume and keyhole plasma. The displacement platform was used to control the welding direction and the spatial position of 304 stainless steel. HWS30.1320-09.02.06 T welding head was provided with four coaxial powder feeding nozzles. PSA400-11-X precision displacement platform and SC300-2B controller were selected in the experiment to control the spatial position of 304 stainless steel. Meanwhile, the spatial positions of welding head, high-speed camera, pyrometer and spectrometer were fixed.

2.2. Experimental method and materials

In this experiment, the exposure time and frame rate of the high-speed camera were set as 20 μ s and 10000 μ s respectively. The displacement platform was used to control the spatial position of 304 stainless steel and the welding speed was set as 20 mm/s. The displacement platform has three motion states: accelerated motion, uniform motion and decelerated motion. Therefore, the uniform movement time of the displacement platform was set to be longer than the welding time, so as to ensure the uniform movement of 304 stainless steel in the welding process. In addition, The power of fiber laser was set as 2 kW and the welding time was set as 1 s. The powder feeding rate was set as 0, 5, 10, 15, 20, 25 g/min. The powder feeding rate of traditional laser welding was 0 g/min. The experimental materials were 304 stainless

steel and Fe50 powder, and their chemical compositions were listed in Table 1.

In order to study the effect of powder feeding rate on surface heat and mass transfer behavior. The surface morphology of the keyhole was photographed by the method of side-axis shooting. As shown in Fig. 1, the center of the light spot I of the spectrometer coincided with the center of the laser spot, and the angle between the optical axis of the spectrometer and the horizontal plane was set as 45°. The distance between the light spot II of the colorimetric pyrometer and the laser spot was set as 2 mm. In order to study the internal heat and mass transfer behavior. The “sandwich” device was used to observe the heat and mass transfer behavior inside the keyhole in this experiment, as shown in Fig. 2. The optical axis of the high-speed camera was perpendicular to the side of the welding base material, and the keyhole profile morphology was photographed through GG17 glass. The light spot I of the spectrometer was located 1 mm directly below the laser spot.

2.3. Spectroscopic method

A. Electron temperature measurement

The electron temperatures of plasma plume and keyhole plasma were calculated by using the Boltzmann diagram method. Some of the parameters came from the NIST database. The formula of electron

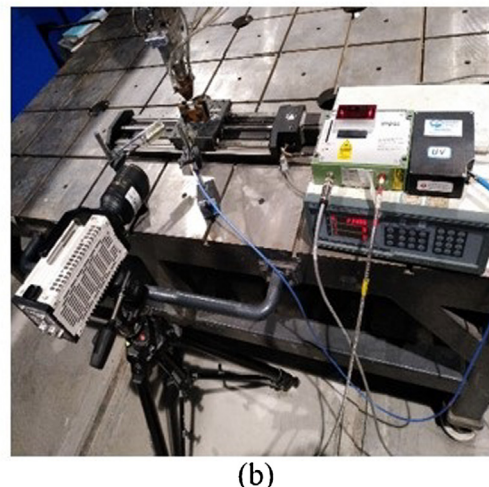
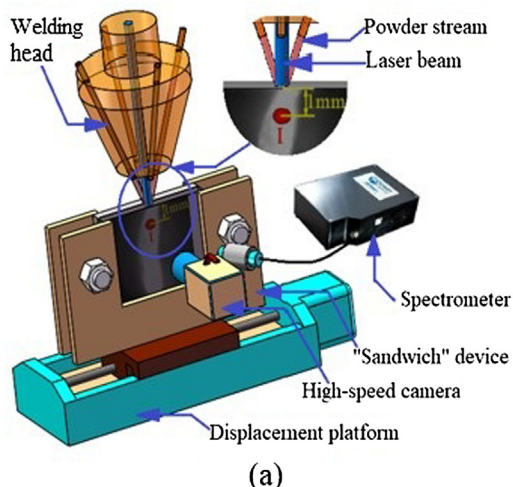


Fig. 2. Internal heat and mass transfer behavior observation platform (a) schematic illustration of experimental set-up, (b) real experimental set-up.

Table 1
Chemical composition (wt%).

Element	C	Cr	Mn	B	Ni	P	Si	S	N	Fe
SUS304 steel	0.08	18–20	2.0	–	8.0–10.5	0.035	1.1	0.03	0.1	Bal
Fe50 powder	0.6–1.0	16–18	–	3–4	12–15	–	3–4.3	–	–	Bal

temperature is as follows [5].

$$\ln \left[\frac{\lambda_{mn} I_{mn}}{g_m A_{mn}} \right] = \ln \left[\frac{N(T_{ex})}{U(T_{ex})} \right] - \frac{E_m}{kT_{ex}} \quad (1)$$

Here λ_{mn} is the line wavelength (nm). I_{mn} is the intensity of light. A_{mn} is the spontaneous emission probability of the transition (10^8 s^{-1}). g_m and E_m are the statistical weight and the energy of the upper level (eV), respectively. $N(T_{ex})$ is the total density of neutral atoms or ions of a specific species. $U(T_{ex})$ is the partition function of this species. $\ln[N(T_{ex})/U(T_{ex})]$ is a constant for a given temperature. k is the Boltzmann constant, $k = 8.625 \times 10^{-5} \text{ (eV/K)}$. Plotting $\ln[\lambda_{mn} I_{mn}/g_m A_{mn}]$ versus E_m for the various spectral lines observed yields a straight line whose slope $-1/kT_{ex}$ provides a measurement of the excitation temperature (K).

B. Electron density measurement

For the electron density of plasma plume, Lorentz fitting was used to obtain the full-width at half-maximum (FWHM) of iron spectral line with a wavelength of 429.31 nm. The electron impact parameter of the spectral line of iron with a wavelength of 429.31 nm was mentioned in [23]. The Stark broadening formula is as follows [6].

$$\Delta\lambda_{1/2} = 2\omega \frac{N_e}{10^{22}} \quad (2)$$

where ω is the electron impact parameter. N_e is the electron density (m^{-3}). $\Delta\lambda_{1/2}$ is the FWHM (nm).

The spectrum of keyhole plasma was different from that of plasma plume. Jay mentioned that Saha equation can be used to calculate electron density of keyhole plasma. Electron density was expressed as a function of the electron temperature in terms of the relative strengths of the two spectral lines. One in Fe (II) and one in Fe (I). The parameters of the formula came from the NIST database and were listed in Table 2. The Saha equation is as follows [2].

$$N_e = c_1 \times 10^6 \cdot T_e^{3/2} \frac{A_{mn}^+ g_m^+ \lambda^0 \varepsilon^0}{A_{mn}^0 g_m^0 \lambda^+ \varepsilon^+} \exp \left[-\frac{V + E_m^+ - E_m^0}{kT_e} \right] \quad (3)$$

where the superscript ‘+’ represents the ionized atom and superscript ‘0’ represents the neutral atom; V is the ionization energy, which is equal to 0.7904 eV for Fe; c_1 is a constant, which value is 2.898×10^{16} . N_e is the electron density of the keyhole plasma (m^{-3}).

C. Plasma absorption mechanism

The main way of absorbing laser energy by plasma is inverse bremsstrahlung absorption (IBA). The calculation formula of inverse bremsstrahlung absorption coefficient is as follows [7].

$$K_v = \frac{Z^2 e^6 N_e N_i \ln \Lambda}{12\pi^{7/2} c^3 \varepsilon_0 [2m_e kT]^{3/2}} \lambda^2 \quad (4)$$

where N_e (m^{-3}) and N_i (m^{-3}) are, respectively, the electron densities and ion densities in a plasma; $\ln \Lambda$ is the coulomb logarithm; Z is ionic valence, $Z = 1$; λ is the laser wavelength; The laser wavelength of YLS-5000 fiber laser is 1070 (nm); ε_0 is vacuum dielectric constant,

Table 2
Data for Saha equation.

λ (nm)	E_m (eV)	A_{mn} (10^8 s^{-1})	g_m
357.54 ⁰	6.49	0.31	5
366.70 ⁺	13.23	0.29	6

$\varepsilon_0 = 8.85 \times 10^{-12} \text{ (F/m)}$; c , e , m_e and k are, respectively, the velocity of light $3 \times 10^8 \text{ (m/s)}$, the electronic charge $1.602 \times 10^{-19} \text{ (C)}$, the electron mass $9.109 \times 10^{-31} \text{ (kg)}$ and Boltzmann constant $1.38 \times 10^{-23} \text{ (J/K)}$. The coulomb logarithm calculation formula is as follows.

$$\ln \Lambda = \ln \frac{12\pi [\varepsilon_0 kT_e]^{3/2}}{Z^2 e^3 N_e^{1/2}} \quad (5)$$

3. Result and discussion

3.1. Electron temperature

The electron temperature calculated by Boltzmann diagram method from the spectrum during FPLW under different powder feeding rate. As shown in Fig. 3, when the powder feeding rate increased from 0 g/min to 25 g/min, the electron temperature of plasma plume first increased by 2% and then decreased by 7%. When the powder feeding rate was 10 g/min, the electron temperature reached the maximum value; the electron temperature of keyhole plasma increased by 1% and then decreased by 5%. When the powder feeding rate was 5 g/min, the electron temperature reached the maximum value.

The reasons for the above-mentioned phenomenon are as follows. According to Eq. (1), the electron temperature of the plasma is proportional to the light intensity of the plasma. The formula of light intensity is as follows.

$$I_{mn} = \frac{Nh\nu}{At} \quad (6)$$

where I_{mn} is the light intensity of the plasma. ν is the frequency of light in the plasma. A is the area of the irradiated region. N is the total number of photons in the irradiated region within the time interval t . In this experiment, the measuring region of the spectrometer was the irradiated region. Because the spatial position of the spectrometer was fixed, the area of the measuring region of the spectrometer did not change. Therefore, the total number of photons emitted by the plasma in the measurement area was the main influencing factor of light intensity. Furthermore, the plasma emitted photons in the following way. When the electrons got enough energy from the laser, they completely broke away from the atoms to form free electrons. At the same time, electrons might have bonded with ions as they moved. In the process of

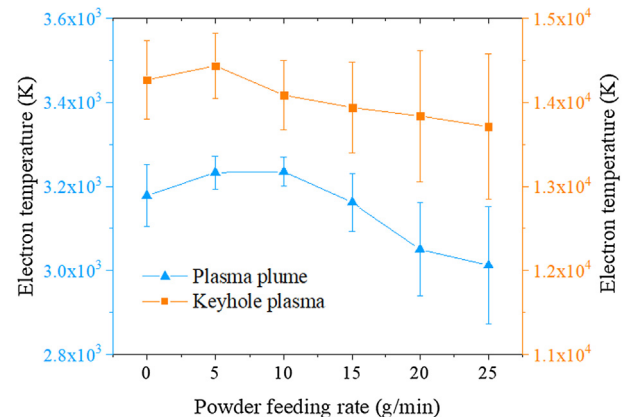


Fig. 3. The electron temperature versus powder feeding rate.

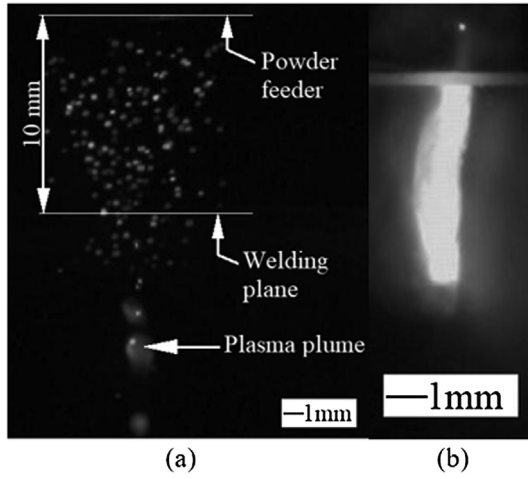


Fig. 4. The process of powder forming plasma.

combining electrons with ions, the energy of the electrons was released by emitting photons. The formula for calculating the number of photons is as follows.

$$N = \frac{E}{h\nu} \quad (7)$$

where E is the total energy of the photons. h is Planck constant. ν is the frequency of light. Metal vapors absorbed energy from the laser beam and ionized to form plasma. A large number of electrons in the plasma broke away from the nucleus to form free electrons. The kinetic energy of the free electrons was proportional to the energy of the laser beam absorbed by the plasma. Therefore, the number of photons emitted by the plasma was proportional to the energy the plasma absorbs from the laser beam. Hence, the powder feeding rate affected the electron temperature by affecting the absorption of laser energy by the plasma. In order to further study the influence of powder feeding rate on electron temperature, the process of powder particles forming metal vapor and plasma in laser-powder coupling experiment and laser welding experiment was photographed. The laser-powder coupling experiment is shown in Fig. 4(a). When the laser power was set as 2Kw, the powder particles were irradiated by the laser and moved about 5–6 mm below the welding plane to form plasma plume. Therefore, the energy of the laser beam was insufficient to vaporize the powder particles above the welding plane. As shown in Fig. 4(b), in the FPLW process, the plasma formed by the welding base material and the laser transmitted energy to the powder particles, which caused the powder particles to form the plasma plume above the welding plane. Hence, the plasma formed by the welding base material provided energy for the process of vaporization of the powder particles. With the increase of powder feeding rate from 0 g/min to 5 g/min, the plasma formed by welding base material and the laser transmitted energy to powder particles to make powder particles vaporize, which resulted in the increase of metal vapor density and electron density. Therefore, the plasma and metal vapor absorbed more laser energy. Furthermore, the light intensity of the plasma increased, which led to the increase of the electron temperature of the plasma. However, with the increase of powder feeding rate from 5 g/min to 25 g/min, because the laser power in this experiment was a fixed value, and the average energy obtained by each powder particle decreased, which made it difficult for the powder particles to completely vaporize. When liquid or solid powder particles were irradiated by the laser beam, the surface of the liquid or solid powder particles vaporized to form the plasma. Therefore, compared with the completely vaporized powder particle, the partially vaporized powder particle formed less metal vapor and plasma and absorbed less laser energy. Therefore, the light intensity of plasma decreased, which led to the decrease of electron temperature of plasma plume and

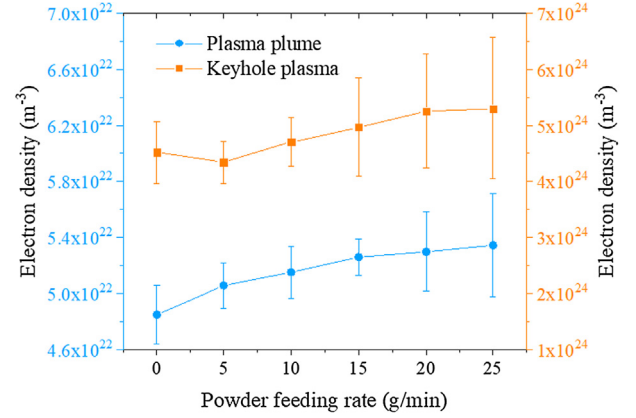


Fig. 5. The electron density versus powder feeding rate.

keyhole plasma.

3.2. Electron density

As shown in Fig. 5, with the increase of powder feeding rate from 0 g/min to 25 g/min the electron density of plasma plume increased by 13%. With the increase of powder feeding rate from 0 g/min to 5 g/min, the electron density of the keyhole plasma decreased by 4%. With the increase of powder feeding rate from 5 g/min to 25 g/min, the electron density of keyhole plasma increased by 22%.

The reasons for the variation of electron density are as follows. For plasma plume, according to Eq. (2), the variation of FWHM determines the variation of electron density. The irregular collision of a large number of particles in plasma plume was the main cause of spectral line broadening. Therefore, particle collision in plasma plume was the main cause of electron density variation in plasma plume. For keyhole plasma, according to Eq. (3), the variation of electron temperature was the main reason for the variation of electron density in keyhole plasma.

The coulomb collision was the main form of interaction collision between charged particles of plasma plume. The spectral line broadening of plasma plume was proportional to the coulomb collision frequency of charged particles of plasma plume, $\Delta\lambda_{1/2} \propto f_c$. The formula of the frequency of coulomb collisions is as follows

$$f_c = 2.9 \times 10^{-12} \frac{N_e \ln \Lambda}{T_e^{3/2}} \quad (8)$$

where N_e (m⁻³) and T_e (K) are, respectively, the electron density and the electron temperature. $\ln \Lambda$ is the coulomb logarithm.

As shown in Fig. 6, with the increase of powder feeding rate, the collision frequency of charged particles of plasma plume increased, which led to the increase of FWHM of the spectrum of plasma plume. Therefore, the electron density of plasma plume increased with

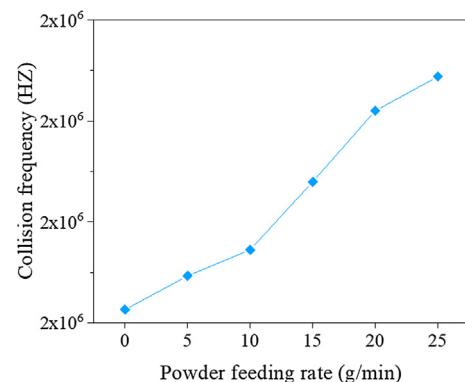


Fig. 6. Charged particles collision frequency versus powder feeding rate.

the increase of powder feeding rate.

For keyhole plasma, according to Eq. (3), the electron density of keyhole plasma was inversely proportional to the electron temperature. Therefore, with the increase of powder feeding rate from 0 g/min to 5 g/min, the increase of electron temperature led to the decrease of electron density. With the increase of powder feeding rate from 5 g/min to 25 g/min, the decrease of electron temperature led to the increase of electron density.

3.3. Keyhole morphology and plasma

A. Keyhole area and plasma plume pressure

In order to find the correlation between the keyhole area and the pressure of plasma plume at the keyhole opening, the pressure calculation of plasma plume was derived as follows. According to formula (9), the plasma plume was in thermodynamic equilibrium. In plasma plume and metal vapor, the motion of uncharged microscopic particle obeys Maxwell velocity distribution. Furthermore, the pressure of plasma plume can be calculated by the equation of state of ideal gas.

$$N_e \geq 1.6 \times 10^{18} T_e^{1/2} (\Delta E)^3 \quad (9)$$

where ΔE is the largest energy gap of the atomic level system (eV). N_e is electron density (m^{-3}).

The microscopic particles in plasma plume included free electrons, iron ions and iron atoms. Therefore, according to the ideal gas state equation, the formula for calculating plasma pressure was derived as follows.

$$P = N_0 \cdot k \cdot T_e \quad (10)$$

where N_0 are the atomic density of plasma plume (m^{-3}). k is the Boltzmann constant 1.38×10^{-23} (J/K). T_e is the electron temperature of plasma plume (K). In the state of local thermodynamic equilibrium, the iron atom ionized and lost an electron. The electron density was equal to the ion density. Then, the Saha equation was used to obtain the atomic density in the plasma plume [6].

$$\frac{N_e N_i}{N_0} = \frac{2g_i [2\pi m_e k T_e]^{3/2}}{g_0 h^3} \exp \left[\frac{-E_i}{k T_e} \right] \quad (11)$$

where m_e is the electron mass; E_i is the ionization potential for the neutral atoms; $g_{i,0}$ are the ground state statistical weight of ions and neutral atoms respectively; h is the Planck constant, $h = 6.626 \times 10^{-34}$ (J·s). Furthermore, according to formula (10) and formula (11), the plasma pressure calculation formula is as follows.

$$P = \left(N_e^2 \frac{g_0}{2g_i} \frac{h^3}{[2\pi m_e k T_e]^{3/2}} \exp \left[\frac{E_i}{k T_e} \right] \right) \cdot k \cdot T_e \quad (12)$$

As shown in Fig. 7, with the increase of powder feeding rate from 0 g/min to 5 g/min, the keyhole area decreased by 4%. With the

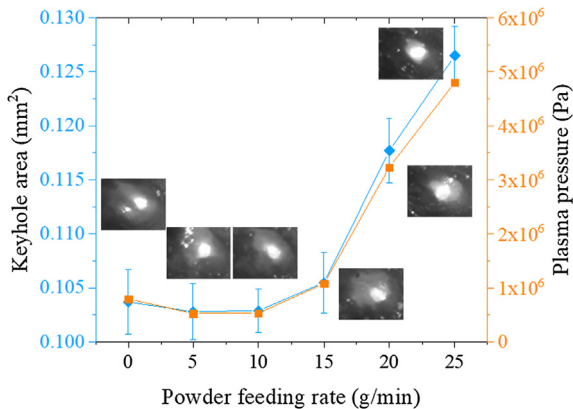


Fig. 7. Keyhole area and plasma plume pressure versus powder feeding rate.

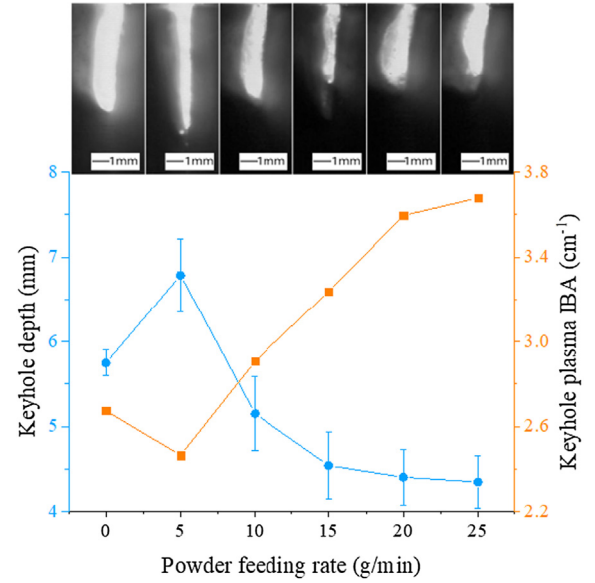


Fig. 8. Keyhole depth and IBA versus powder feeding rate.

increase of powder feeding rate from 5 g/min to 25 g/min, the keyhole area gradually increased by 23%. According to formula (12), the pressure of plasma plume was obtained, as shown in Fig. 7. The variation of keyhole area was similar to that of plasma plume pressure. During the filler powder laser welding, the keyhole continuously sprayed plasma to the ambient air. The plasma plume pressure was positively correlated with the plasma pressure at the keyhole opening. Therefore, with the increase of powder feeding rate, the pressure of plasma plume at the keyhole opening increased, resulting in the increase of keyhole area and plasma plume pressure.

B. Keyhole depth and keyhole plasma IBA

As shown in Fig. 8, with the powder feeding rate increased from 0 g/min to 5 g/min, the keyhole depth increased by 18%. With the powder feeding rate increased from 5 g/min to 25 g/min, the keyhole depth decreased by 36%. This phenomenon can be explained by the electron density and the inverse bremsstrahlung absorption of the keyhole plasma. As shown in Fig. 5, when the electron density of the keyhole plasma at the top of the keyhole increased, the inverse bremsstrahlung absorption coefficient of the keyhole plasma increased, which meant that the keyhole plasma absorbed more laser energy. The keyhole plasma at the top of the keyhole absorbed the laser energy and transmitted the energy to the keyhole wall at the top of keyhole, which blocked the laser energy absorbed by the bottom wall of the keyhole. As shown in Fig. 9, with the powder feeding rate increased from 0 g/min to 5 g/min, the inverse bremsstrahlung absorption coefficient of the

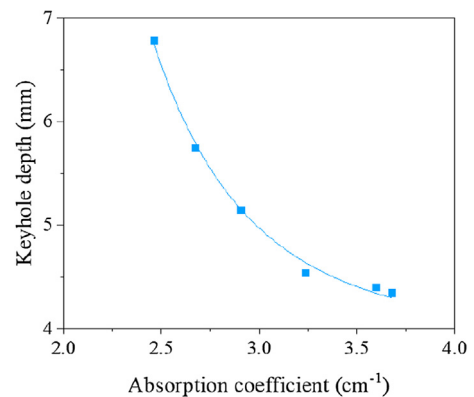


Fig. 9. Correlation curve of keyhole depth and keyhole plasma IBA.

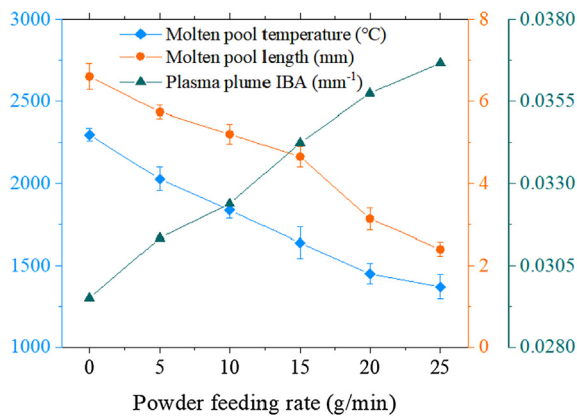


Fig. 10. Correlation between thermal and mass transfer behavior of molten pool and plasma plume IBA.

keyhole plasma decreased. The laser energy absorbed by the wall at the bottom of the keyhole increased, which led to the increase of the keyhole depth. With the increase of powder feeding rate from 5 g/min to 25 g/min, the inverse bremsstrahlung absorption coefficient of keyhole plasma increased. The laser energy absorbed by the wall at the bottom of the keyhole decreased, which led to the reduction of the keyhole depth. Therefore, in this paper, 5 g/min is the critical point of powder feeding rate for 304 stainless steel welding base material. As shown in Fig. 10(c), the keyhole depth is inversely proportional to the inverse bremsstrahlung absorption coefficient of keyhole plasma. Therefore, in the process of filler powder laser welding, the keyhole depth and fusion depth can be improved by selecting critical points of powder feeding rate as laser welding process parameters.

3.4. Molten pool temperature and length

In the FPLW process, with the increase of the powder feeding rate, the surface temperature of the molten pool was measured by using the colorimetric pyrometer. As shown in Fig. 10, with the powder feeding rate increased from 0 g/min to 25 g/min, the surface temperature of the molten pool decreased by 930 °C. The length of the molten pool decreased with the increase of the powder feeding rate. The reasons for above-mentioned phenomenon was described as follows. On the one hand, as shown in Figs. 5 and 10, electron density and inverse bremsstrahlung absorption (IBA) coefficient of plasma plume increased with the increase of powder feeding rate. When the electron density of the plasma plume above the keyhole increased, the plasma plume can not only absorb the energy of the laser by means of IBA, but also blocked the energy transfer of the laser by means of scattering and reflecting. The plasma plume absorbed the laser energy and contacted with the ambient air, releasing heat through thermal radiation and convection. Therefore, with the increase of the inverse bremsstrahlung absorption coefficient of plasma plume, the energy absorption of the keyhole decreased. Furthermore, the keyhole transferred less heat to the molten pool, resulting in lower surface temperature of the molten pool. On the other hand, with the increase of powder feeding rate, the number of powder particles sprayed into the molten pool increases. The heat transferred from the molten pool to more and more powder particles resulted in the lower temperature of the molten pool.

4. Conclusions

In this paper, experimental platforms were set up to study the influence of powder feeding rate on the heat and mass transfer behavior during filler powder laser welding of 304 stainless steel. The main conclusions were summarized as follows.

- (1) Compared with the keyhole depth of traditional laser welding, when the powder feeding rate was 5 g/min, the electron density and inverse bremsstrahlung absorption coefficient of keyhole plasma in the process of filler powder laser welding were reduced by 4% and 8% respectively, which resulted in the increase of keyhole depth by 18%. In the process of filler powder laser welding, the critical value of powder feeding rate is 5 g/min. When the powder feeding rate approached the critical value, the keyhole depth and fusion depth increased.
- (2) As the powder feeding rate increased from the critical value 5 g/min, electron temperature and electron density decreased and increased respectively in plasma plume and keyhole plasma. The powder particles affected the electron temperature and the electron density respectively by affecting the number of photons emitted by the plasma and the collision frequency of charged particles in the plasma.
- (3) When the powder feeding rate exceeded the critical value, the plasma pressure at the keyhole opening and the IBA coefficient of the plasma at the top of the keyhole increased, resulting in the increase of the keyhole area and the decrease of the keyhole depth. The powder particles affected plasma pressure by affecting electron temperature and electron density. The increase of plasma pressure at the keyhole opening led to the increase of keyhole area. The IBA coefficient of the keyhole plasma at the top of the keyhole increased, which blocked the laser beam irradiating the wall at the bottom of the keyhole and caused the decrease of keyhole depth.
- (4) There are two reasons for the decrease of the surface temperature of the molten pool. On the one hand, with the increase of the powder feeding rate, the electron density and the IBA of plasma plume increased. Plasma plume refracted and reflected laser beams. The plasma plume absorbed the laser energy, which caused the keyhole and molten pool to absorb less laser energy by the thermal radiation and convection of plasma plume with the ambient air. On the other hand, more and more powder particles were sprayed into the molten pool and absorbed heat from the molten pool, which caused the surface temperature of the molten pool to decrease.

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